

EDUCATION

B.Eng. Mechanical Engineering

Polytechnique Montréal

2020 — 2026 · Graduated

RELEVANT COURSES

- Numerical Methods — FEM, FDM, COMSOL
- Structural Analysis & Mechanics of Materials
- Fluid Mechanics & Thermodynamics
- Applied AI in Engineering (ML, Neural Networks)
- System Dynamics & Control
- Computer-Aided Design — GD&T

SKILLS

SIMULATION / FEM

COMSOL Advanced

FEM / FDM Advanced

CFD Advanced

ANSYS Intermediate

Simulia Intermediate

OpenFOAM Intermediate

CAD & DESIGN

SolidWorks Advanced

CATIA Advanced

Inventor Intermediate

Enovia Intermediate

ML / AI

Gaussian Process Expert

scikit-learn Advanced

PyTorch Advanced

TensorFlow Intermediate

PROGRAMMING

Python Expert

MATLAB Advanced

FastAPI Intermediate

Git Intermediate

LANGUAGES

French Native language

English B2 Level (Upper Intermediate)

PROFILE

Mechanical Engineer · Polytechnique Montréal (2026) — specialized in **FEM** (COMSOL, ANSYS) and **machine learning applied to physical simulation**. Built a **Gaussian Process** surrogate model (**$R^2=0.9999$** , **100,000×** speedup over FEM, 500 COMSOL simulations) and an audio CNN pipeline (**93.34%** accuracy). End-to-end ownership from specifications to tested prototype — Neptronic internship (HVAC, **GD&T**) and 4 delivered technical projects. Proficient in CAD (SolidWorks, CATIA), CFD and production-ready API deployment.

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern

May 2023 — Aug 2024

Neptronic · Montréal, QC · HVAC Systems & Industrial Humidification

- Modeled a 20×20 cm HVAC air inlet damper (Autodesk Inventor) — airflow compliant with specifications on **first iteration**, zero revision cycles.
- Developed a **PCB control system for the humidifier**: circuit schematic, CFD analysis and supplier data integration.
- Designed a quality measurement fixture with **GD&T** dimensioning for the production test line.
- CAD design (Autodesk Inventor) of an actuator crank arm for air inlet valve (target: **50 in·lbs**) and auxiliary switch mounting bracket — torque achieved: **40 in·lbs** (80% of target, limited by assembly constraints), bracket fit validated in production.

TECHNICAL PROJECTS

TPS Thermal Surrogate

2025

- Built a Finite Difference (FDM) → **Finite Element (FEM)** → Machine Learning pipeline for thermal modeling of an atmospheric reentry heat shield.
- Gaussian Process surrogate model (**500** COMSOL simulations): **$R^2=0.9999$** , **0.03%** error, **100,000×** speedup over FEM.
- Trained a full-field MLP $T(x,y,t)$ on a 21×21×181 grid: **$R^2=0.9875$** , **600×** speedup.
- Deployed an interactive Streamlit app with real-time GO/NO-GO heat map for field engineers.

COMSOL

Python

scikit-learn

Gaussian Process

MLP

FEM

Streamlit

HarmoniQ — Electrical Grid Simulator

2024 — 2025

- Built a modular Python package simulating optimal dispatch of **35+** power plants — replicating the real Hydro-Québec grid topology.
- Integrated **30+** geospatial watersheds, **35+** power plants and MRC transmission topology.
- Modeled **5** energy sources (hydro, wind, solar, nuclear, thermal) and power flows via PyPSA.
- Exposed a Flask REST API for dynamic optimal dispatch queries and network flow visualization.

Python

PyPSA

SQLAlchemy

Pandas

pvlb

Flask

NeuroSound — Acoustic Classifier

2023 — 2024

- Designed **2** independent CNN pipelines — ESC-50 dataset, **50** classes, **2,000** samples.
- EfficientNetB3 + multi-head attention → **93.34%** accuracy; YAMNet (AudioSet) fine-tuned → **88.5%**.
- Automated audio preprocessing: MFCC, mel-spectrogram, augmentation, normalization via NumPy/librosa.
- Production-ready FastAPI — async SQLAlchemy, Docker, lifecycle handlers.

TensorFlow

EfficientNetB3

YAMNet

FastAPI

Docker

SQLite

POL-E — Bluetooth-Teleoperated Agricultural Robot

2023

- Designed a Bluetooth robot (**1.2 kg**, 6 V DC) in SolidWorks/CATIA — from CAD modeling to physical assembly.
- Validated PLA fork by **structural analysis**: $\sigma_{max} = 25.35 \text{ MPa}$, $F_s = 1.95$.
- Programmed Arduino C++ control system: Bluetooth teleoperation, DC motor and actuator management.
- 1st payload/time ratio** — POL-E 2023 Competition: **400 g** transported in **45 seconds**.

SolidWorks

CATIA

Arduino

C++

Structural Analysis

Bluetooth